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Wave aberration corrections in PSF ellipticity measurements of astronomical telescopes using a multi-objective optimization

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The weak gravitational lensing (WGL) produces a shear effect on the observed galactic ellipticity that is much smaller than the endogenous ellipticity of the galaxy itself. Achieving such a high-level astronomical observation requires the superior performance of telescopes. To ensure the optical properties of telescopes to be competent in WGL detections, it is very necessary to measure point spread function (PSF) ellipticity of telescopes in labs. In this paper, a 2 m off-axis telescope that would be used to detect WGL in space is analyzed and studied. A collimator whose aperture is 2 m has been built to measure PSF ellipticity of the telescope. The wave aberrations of the collimator are roughly equal to those of the telescope, so they are important systematical errors and must be removed. However, it is difficult to precisely measure the wave aberrations of optical systems that have large apertures and long focal lengths. In addition, a 2 m flat mirror, which is indispensable to measure wave aberrations of optical systems, has significant surface errors. In this paper, a multi-objective optimization method is proposed to eliminate the effects of wave aberrations on PSF ellipticity measurements of the telescope. By constructing an equivalent model, the wave aberrations from collimators and flat mirrors can be corrected so that PSF ellipticity measurement error is reduced to within 0.01. Measurement accuracy of PSF ellipticity of the telescopes can be improved significantly. © 2024 Optica Publishing Group

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1. INTRODUCTION

Weak gravitational lensing (WGL) is the slight deformation of light from background galaxies by the gravitational field formed by dark matter and dark energy. Hence, WGL is a good probe for studying cosmic dark matter and dark energy [1]. However, the WGL causes very small changes in the galaxy's ellipticity, called shear deformations, which have much smaller values than the galaxy's ellipticity [2,3]. To meet the observational requirements of the WGL, several powerful astronomical telescopes have been proposed, including the Euclid Space Telescope [4,5], Nancy Grace Roman Space Telescope [6], Vera C. Rubin Observatory Legacy Survey of Space and Time [7], and the Chinese Space Station Telescope [8]. Space telescopes avoid the effects of the atmosphere [7] so that it is easier for them to achieve diffraction-limited image quality. However, space telescopes require more stringent error control compared to ground-based telescopes [9–12]. Since the shapes of galaxies must be convolved by the point spread function (PSF) of telescopes before we observe them, optical properties of telescopes are important systematic errors in WGL detections [13].

The accuracy of WGL detection can be improved by a comprehensive understanding of the optical properties of telescopes [14]. Zhang *et al*. [15] designed an off-axis space telescope to detect WGL, in which tight constraints are required on the spatial stability of unobstructed telescopes and PSF ellipticity is used to evaluate the image quality. Luo *et al*. [16] analyzed the effect of polarization aberration on the ellipticity of the telescope PSF using polarization ray tracing. Zeng *et al*. [17] investigated the effect of geometric aberration on the PSF ellipticity, and Monte Carlo simulations of the PSF ellipticity were used to guide the accurate calibration of off-axis telescopes. Chang *et al*. [10] simulated systematic errors such as reflector misalignment, wavelength response of optical elements, occlusion of telescope components, and scattered rays.

PSF ellipticity is an optical property of telescopes and depends on many factors, such as optical design, manufacturing errors, optical alignment, and so on. Hence, there may be big differences between the actual PSF ellipticity of telescopes and the ellipticity obtained in the design process. In order to ensure the ability of telescopes to accurately measure WGL, it is necessary to measure PSF ellipticity of telescopes in labs.

Currently, the PSF measurements of telescopes have been studied extensively. Liaudat *et al*. proposed the multi-CCD PSF modeling method applied to telescope imaging observations of galaxies [18]. The team of the Euclid mission used a Shack–Hartmann sensor to measure wave aberration and evaluated the image quality by comparing it with the PSF [19].

However, the error sources that dominate PSF ellipticity measurements for telescopes are very different from those in PSF measurements. For example, non-rotationally symmetrical wave aberrations have greater impacts on PSF ellipticity. More importantly, the requirements of PSF ellipticity for errors are more stringent. Hence, measuring PSF ellipticity of telescopes in the labs presents many challenges. Fan *et al.* [20] have measured the PSF ellipticity of an off-axis reflective telescope by star point imaging under laboratory conditions. Many errors including centroid positioning errors and defocusing of the star target have been analyzed. As we all know, collimators are indispensable to measure PSFs of telescopes on the ground. The aperture of collimators should not be smaller than that of the telescope that is to be measured. Hence, a collimator whose aperture is 2 m has been built in our labs and would be used to measure PSF ellipticity of an off-axis telescope that is planned to detect WGL in space. However, the wave aberrations of the collimator are roughly equal to those of the telescope and would become important error sources. As a result, it is critical to eliminate the effects of the collimator.

In this paper, a multi-objective optimization method is proposed to eliminate the effects of wave aberrations on PSF ellipticity measurements of the off-axis space telescope. The rest of this paper is arranged as follows. In Section 2, the off-axis space telescope is introduced and the effects of the factors in the wave aberration numerical calculation method and the star point target imaging method on PSF ellipticity measurements are analyzed. The evaluation functions that are used for iteration and the flow of the algorithm are given in Section 3. In Section 4, the optimized compensation results are shown to demonstrate the feasibility and robustness of the method proposed in this paper. A discussion and conclusion are presented in Section 5.

2. PRINCIPLE AND METHOD

Currently, wave aberration numerical calculation [17] and star point target imaging [20] are the two common methods to obtain a telescope's PSF, which also enable to obtain PSF ellipticity in labs. In this section, a typical off-axis telescope is introduced, and wavefront measurement data of this optical system are used as an example in this paper. A PSF ellipticity calculation model of optical systems containing wave aberrations is provided. Using fast Fourier transform (FFT) on optical pupil function (OPF), the influence of the wave aberrations on PSF ellipticity measurement can be obtained by analyzing the relationship between the Zernike coefficient and PSF ellipticity. For the star point target imaging method, the effects of collimators are analyzed. Advantages and disadvantages of the two PSF ellipticity measurement methods are summarized. Based on this, a multi-objective optimization method for PSF ellipticity measurement of astronomical telescopes is proposed.

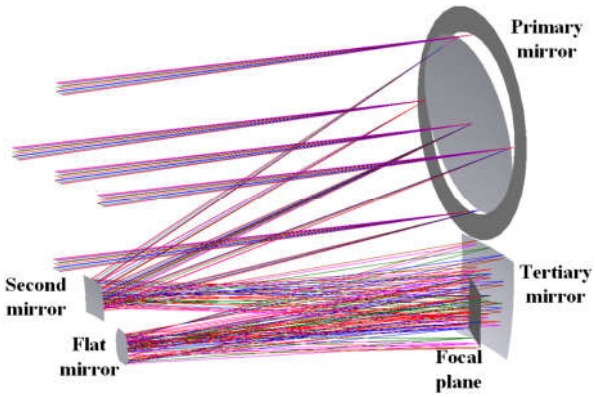


Fig. 1. Typical off-axis telescope layout.

A. Typical Off-Axis Optical System

In this paper, a telescope that would be used to detect WGL is analyzed and shown in Fig. 1. It is a 2 m F/14 Cook three-mirror anastigmat telescope with a $1.1° \times 1°$ field of view (FOV). The average root mean square (RMS) wave aberration is about 40 nm. In order to build a star point target imaging optical system in labs, a collimator is used, whose aperture is 2 m, the focal length is 70 m, the average RMS wave aberration is about 45 nm, and the effective FOV is 2′. The simulation parameters of the telescope's PSF are defined [20], where the grid spacing between samples in the image plane is 0.4 μm, the total number of rays across the transform grid is 1024×1024, and the wavelength is 632.8 nm.

B. PSF Ellipticity Calculation Model for Optical Systems with Aberrations

PSF ellipticity is a parameter that characterizes the shape and spatial distribution of PSFs [21,22]. If $I(x, y)$ indicates the intensity of an image, and $W(x, y)$ indicates a weight function, the center of the image is given by the weighted first-order moments divided by the total intensity:

$$\begin{cases} \bar{x} = (1/F) \int x I(x, y) W(x, y) \mathrm{d}x\mathrm{d}y \\ \bar{y} = (1/F) \int y I(x, y) W(x, y) \mathrm{d}x\mathrm{d}y, \end{cases} \tag{1}$$

$$F = \int I(x, y) W_0(x, y) \mathrm{d}x\mathrm{d}y. \tag{2}$$

The quadrupole moment is obtained by the weighted second moments divided by the total intensity:

$$\begin{cases} Q_{11} = (1/F) \int (x - \bar{x})^2 I(x, y) W_0(x, y) \mathrm{d}x\mathrm{d}y \\ Q_{12} = (1/F) \int (x - \bar{x})(y - \bar{y}) I(x, y) W_0(x, y) \mathrm{d}x\mathrm{d}y \\ Q_{22} = (1/F) \int (y - \bar{y})^2 I(x, y) W_0(x, y) \mathrm{d}x\mathrm{d}y \end{cases}. \tag{3}$$

The PSF of an optical system containing aberrations can be obtained using FFT on OPF. The second-order moments, Q_{mn}, are given by an integrated expression for the PSF of the optical system in the image plane:

$$Q_{mn} = \frac{\int \left| fft\{\exp[jk \sum_i A_i Z_i(\xi, \eta)] P(\xi, \eta)\} \right|^2 c_m c_n W(x, y) \mathrm{d}x\mathrm{d}y}{\int \left| fft\{\exp[jk \sum_i A_i Z_i(\xi, \eta)] P(\xi, \eta)\} \right|^2 W(x, y) \mathrm{d}x\mathrm{d}y}, \tag{4}$$

where A_i represents the aberration coefficient, $Z_i(\zeta, \eta)$ represents the Zernike polynomial in the pupil, and $P(\zeta, \eta)$ represents the corresponding OPF. c_m and c_n represent the distance between the x, y coordinate position on the image plane and the intensity center $\bar{x}$ position.

The two component ellipticities e_1 and e_2 are defined by

$$e_1 = \frac{Q_{11} - Q_{22}}{Q_{11} + Q_{22}}, \tag{5}$$

$$e_2 = \frac{2Q_{12}}{Q_{11} + Q_{22}}. \tag{6}$$

The magnitude of ellipticity e is

$$e = \sqrt{e_1^2 + e_2^2}. \tag{7}$$

C. PSF Ellipticity Wave Aberration Numerical Calculation Method

For an optical system, the PSF is determined by the OPF and the wave aberration. According to Eqs. (4)–(7), PSF ellipticity is related to the morphological parameters of the PSF [22]. Obviously, PSF ellipticity can be obtained if both wave aberration and pupil shape are known, which is the principle of FFT on OPF [15].

For the PSF ellipticity wave aberration numerical calculation method, the accuracy of PSF ellipticity calculation directly depends on the wavefront measurement accuracy of the optical system. However, it is difficult to precisely measure the wave aberrations of an optical system that has a large aperture and long focal length. The Zernike coefficient measurement errors increase rapidly with the increase of the aperture and the focal length of the system. We have performed the wavefront measurement experiments on the 2 m off-axis telescope optical system shown in Fig. 1. Under vibration isolation and clean laboratory conditions, the system was measured multiple times using a dynamic 4D interferometer, as shown in Fig. 2(a). The experimental results are shown in Fig. 2(b), where the horizontal coordinates are Fringe Zernike terms, and the vertical coordinates are the mean and the upper and lower deviations (3σ). As shown in Fig. 2(b), the higher-order Zernike coefficients are stable, while the measurement errors of lower-order terms (Z4–Z12) are large. The largest error is in the defocus (Z4), which reaches 0.03λ (632.8 nm), followed by the first-order 0° astigmatism (Z5) and the first-order Y-axis coma (Z8).

In order to obtain the effect of the uncertainty of the Zernike coefficient on PSF ellipticity, we substitute the mean and the upper and lower deviations in Fig. 2 into Eqs. (4)–(7). In Fig. 3, the results are shown using purple and yellow lines, respectively. The PSF ellipticity variation caused by the Zernike coefficient measurement errors varies in each order and is mainly distributed in astigmatism and coma terms. As shown in the blue area, the effect of first-order astigmatism (Z5–Z6) on the PSF ellipticity reaches 0.034, and the first-order coma (Z7–Z8) reaches 0.012. Both of them are larger than the requirement of 0.01 for the system PSF ellipticity measurement accuracy [20]. Obviously, this method cannot meet the requirements for the accuracy of PSF ellipticity calculation.

D. Star Point Target Imaging Method

The star point target imaging method is also commonly used to obtain the PSF ellipticity of optical systems. A collimator is indispensable to obtain a star target on the ground. In our previous work [20], a star point target imaging method was proposed to measure the PSF ellipticity of the telescope. The measurement scheme is shown in Fig. 4(a), which includes a non-coherent illumination system, a pinhole target simulator with a diameter of 50 μm (according to Rayleigh's criterion), a collimator, an off-axis optical system, and an image acquisition system. A detector is combined with a 10X microscope objective to acquire PSF images with the required PSF ellipticity calculation accuracy. Obviously, the collimator would directly affect the PSF of the entire optical path. The wave aberrations of our collimator, whose clear aperture is 2 m, are measured. In order to clarify the effect of introducing a collimator with wave aberrations of 0.03λ (632.8 nm) on the PSF ellipticity measurement error, Monte Carlo simulations were performed, and the results are shown in Fig. 4(b). After the collimator is introduced, the mean variation of PSF ellipticity is 0.081, which is bigger than the required 0.01. Hence, the effect of the collimator's wave aberrations on the PSF ellipticity of the system must be removed.

E. Our New Method

The characteristics of the two methods shown in Sections 2.C and 2.D are summarized and shown in Fig. 5. The PSF ellipticity wave aberration numerical calculation method uses system wavefront measurement data obtained from interferometers. However, for a large-aperture off-axis optical system, the effects of wavefront measurements and flat mirror are inevitable and cannot be ignored. It has been proven that this can have a large impact on PSF ellipticity measurements, as shown in Fig. 3. The star point target imaging method uses a detector to obtain the actual imaging characteristics of optical systems. However, a collimator is indispensable, and the wave aberrations of the collimator become a new source of error. Hence, both the PSF ellipticity wave aberration numerical calculation method and the star point target imaging method cannot achieve the requirements of PSF ellipticity calculation accuracy.

As shown in the central portion of Fig. 5, we propose a multi-objective optimization and compensation method to obtain the PSF ellipticity of a large-size off-axis optical system. The biggest feature of the new method is that both wavefront measurement data and star point target images are involved. There are three critical steps to perform the new method. First, the wave aberrations of the collimator and the off-axis optical system should be measured. Of course, measurement errors are inevitable because of the large aperture and long optical path of the off-axis optical system. In addition, the wave aberrations of the flat mirror matched to the aperture of the off-axis optical system also needs to be removed. Second, the collimator

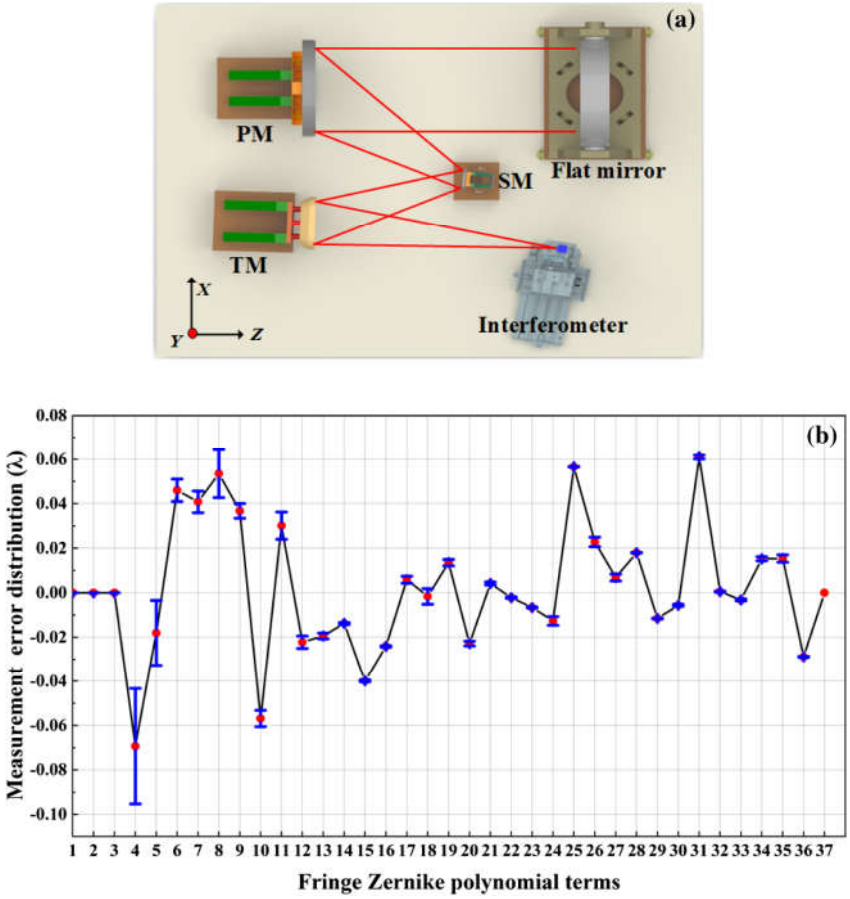


Fig. 2. PSF ellipticity wave aberration numerical calculation method. (a) Off-axis optical system wavefront measurement optical path. (b) Uncertainty of the Zernike coefficient. The red circles indicate the mean values, and the blue lines indicate the upper and lower deviations.

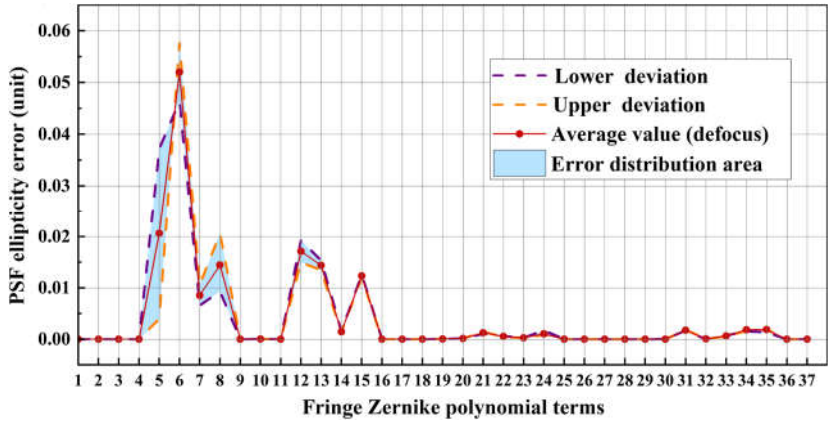


Fig. 3. Effect of each order of the Zernike coefficient uncertainty on PSF ellipticity in the measured values. The orange line indicates the PSF ellipticity calculation error corresponding to the lower deviation, the purple line indicates the PSF ellipticity calculation error corresponding to the upper deviation, and the area enclosed in the middle is the PSF ellipticity calculation error distribution interval.

and the off-axis optical system are aligned in vacuum and the PSF of the whole system is measured, which we will denote as PSF-M. Finally, the measured wave aberrations are input into optical simulation software and the measured PSF-M is set as the optimization objective. The last step is the primary content of the new method proposed in this paper. We will introduce it in detail.

In optical design software, the wave aberrations can be represented by a Zernike phase surface. Hence, both the collimator and off-axis optical system are represented as a combination

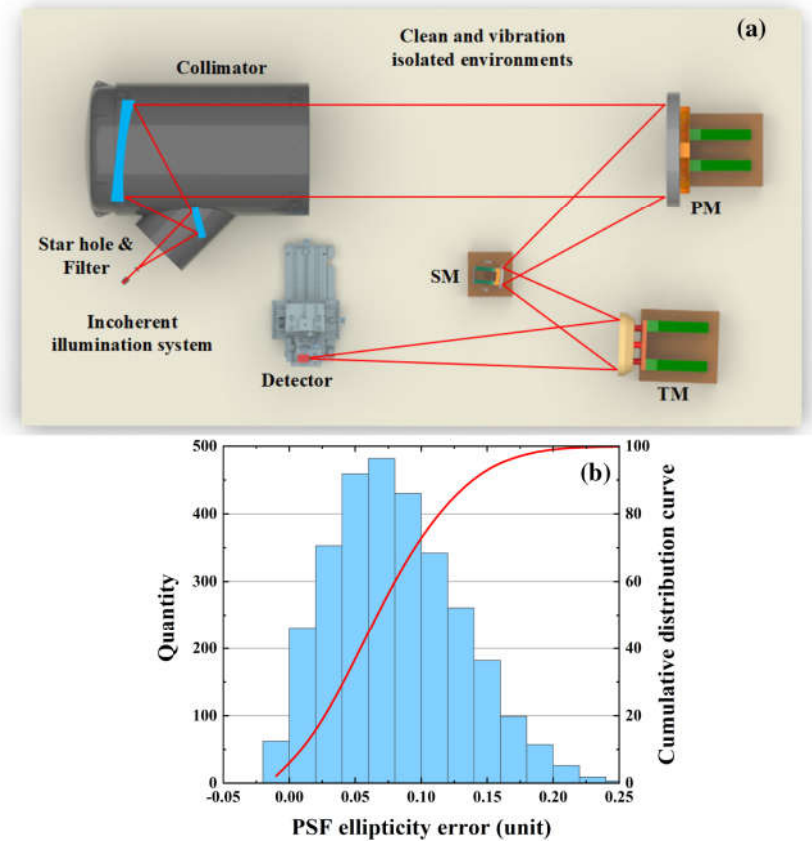


Fig. 4. Star point target imaging method. (a) Schematic diagram of star point target imaging optical path. (b) Effect of collimator wave aberrations on PSF ellipticity. The light blue bars indicate the error distribution intervals and quantities, and the red line is the cumulative distribution curve.

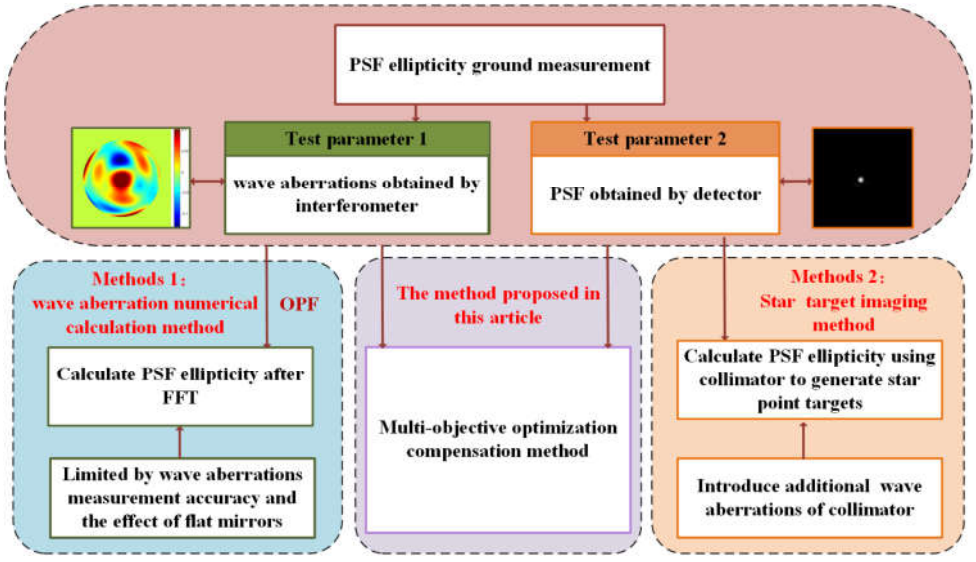


Fig. 5. Summary of current PSF ellipticity detection methods and our new method.

of an ideal thin lens surface and a Zernike phase surface. The two surfaces can be combined as an equivalent lens, as shown in Figs. 6(a) and 6(b). As a result, a star point target imaging optical path is built in optical design software, as shown in Fig. 6(c). Additionally, a "virtual lens" is introduced between the two equivalent lenses, which is a compensating surface determined by Zernike coefficients. The PSF of the star point target imaging optical path, which is built in optical design

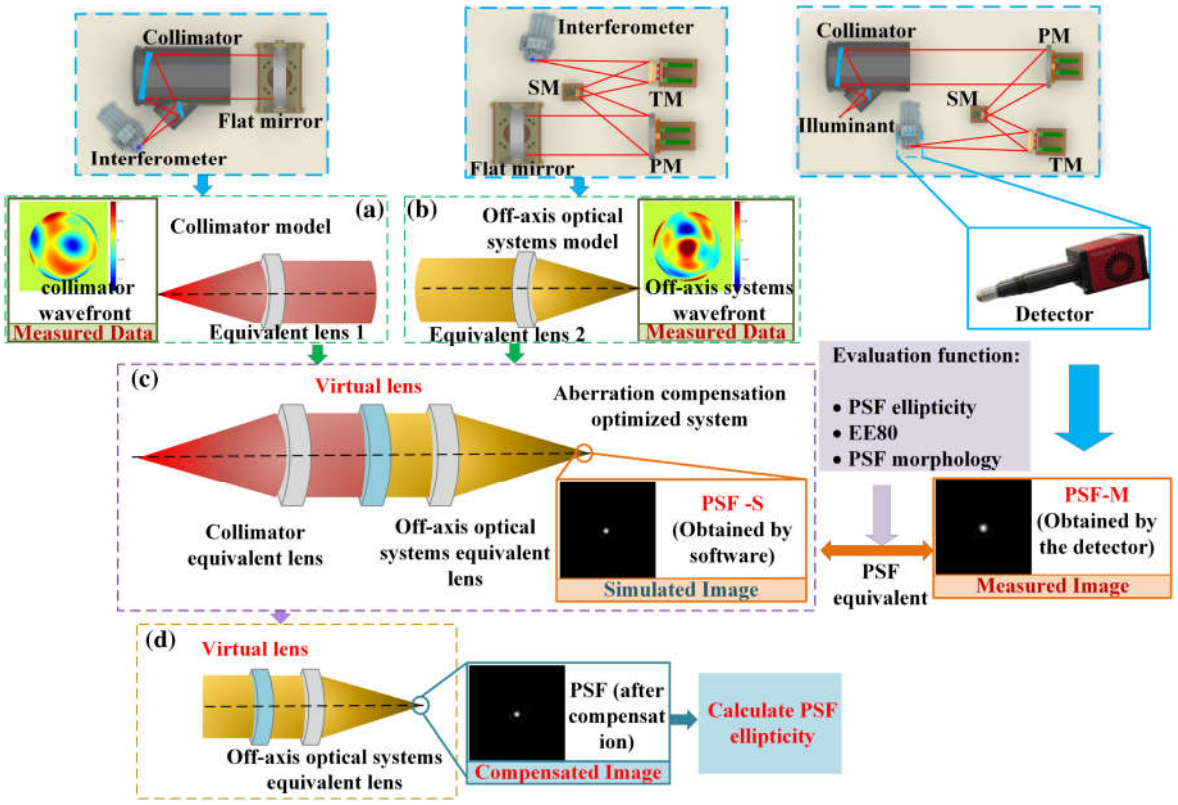


Fig. 6. Optimization model. (a) Collimator equivalent model. (b) Off-axis optical system equivalent model. (c) Aberration compensation optimized system. (d) Off-axis optical system equivalent model with compensation surface.

software, can be obtained and denoted as PSF-S. The PSF-M measured by the detector is used as the optimization objective. The Zernike coefficients of the "virtual lens" are used as the optimization variable so that the PSF-S is continuously varied until the differences between the PSF-S and the PSF-M are small enough. At this point, the combination of the "virtual lens" and the equivalent lens of the off-axis optical system is just the compensation system, as shown in Fig. 6(d). The PSF ellipticity of the compensation system is the PSF ellipticity of the off-axis optical system. In summary, by constructing an equivalent model, the wave aberrations induced by the collimator can be effectively eliminated. A "virtual lens" is used to compensate for wavefront measurement errors and remove the effect of the flat mirror. Lastly, the detailed optimization process, which aims to minimize the differences between the simulated PSF-S and measured PSF-M and would be done by the NSGA-II algorithm, will be discussed in the following section.

3. EVALUATION FUNCTION AND CALCULATION PROCESS

A. Establish the Evaluation Function

The NSGA-II algorithm is one of the most representative algorithms in the multi-objective optimization search algorithm. The construction of evaluation functions is one of the key steps of the NSGA-II algorithm. In this paper, three evaluation functions are given, including PSF ellipticity evaluation function f_e, EE80 radius function f_r, and PSF morphology function f_p. The composite evaluation function $F_c(f_e, f_r, f_p)$ is the sum of the three evaluation functions. Considering NSGA-II has been widely used in dealing with multi-objective evolutionary algorithms problems [23–25], the intricate procedural details of this algorithm will not be introduced in this paper.

1. PSF Ellipticity Evaluation Function f_e

According to the definition of PSF ellipticity, two different PSFs may have the same PSF ellipticity. Therefore, it is not sufficient to solely rely on the PSF ellipticity e as the sole criterion to determine whether the PSFs of the two systems are identical. We introduce the long-axis component based on Eq. (7). The PSF ellipticity evaluation function is shown in Eq. (8):

$$f_e = w_1 |\Delta e| + w_2 |\Delta e_1| , \tag{8}$$

where w_1, w_2 represent the weights, Δe indicates the difference between the PSF ellipticity of the optimized system and the target system, and Δe_1 indicates the long-axis difference.

2. EE80 Radius Function f_r

The f_r refers to the radius of the circular area that contains 80% of the total energy of the PSF [26]. It is a common parameter to describe the characteristics of PSFs. Considering the size of PSF depends on wavelength, 632.8 nm is chosen as the calculated

wavelength in this paper. The evaluation function is given in Eq. (9):

$$f_r = w_3 |r_2 - r_1| = w_3 |\Delta EE|, \tag{9}$$

where w_3 represents the weight, r_1 indicates the EE80 radius of PSF-M, r_2 indicates the EE80 radius of PSF-S, and ΔEE indicates the difference between the EE80 radius of the two PSFs.

3. PSF Morphological Function f_p

The PSF morphological function is introduced to evaluate the morphological differences between the PSF-M and the PSF-S, as shown in Eq. (10):

$$f_p = w_4 \sum_{j=1,k=1}^{n} \left[\mathrm{PSF}_{\mathrm{Opt}}(x_j, y_k) - \mathrm{PSF}_{\mathrm{Dis}}(x_j, y_k)\right]^2, \tag{10}$$

where w_4 represents the weight factor, $\mathrm{PSF}_{\mathrm{Dis}}(x_j, y_k)$ indicates the PSF-M, and $\mathrm{PSF}_{\mathrm{Opt}}(x_j, y_k)$ indicates the PSF-S.

B. Calculation Process

The framework proposed in this paper can be divided into two parts: optimization model training and NSGA-II. The optimization model training can be divided into three steps, as shown in Fig. 7. (1) Acquiring measurement data to construct aberration compensation optimization system models, including star point target image, off-axis system wave aberrations, and collimator wave aberrations. (2) Adding compensation surfaces into the simulation model and generating simulated star point images. Use NSGA-II and the evaluation function to make the difference between PSF-M and PSF-S small enough and obtain a pareto solution. (3) Obtaining the compensation structure and calculating the compensated ellipticity interval. In addition, as shown by the light blue dotted line, the application of NSGA-II in the optimization model is to obtain a surface used to compensate for the measurement error of wave aberrations of the off-axis optical system. The iterative strategies included simulated binary crossover, polynomial variation, and binary tournament sorting methods for NSGA-II. The analysis of specific cases can follow the steps shown in the flowchart and

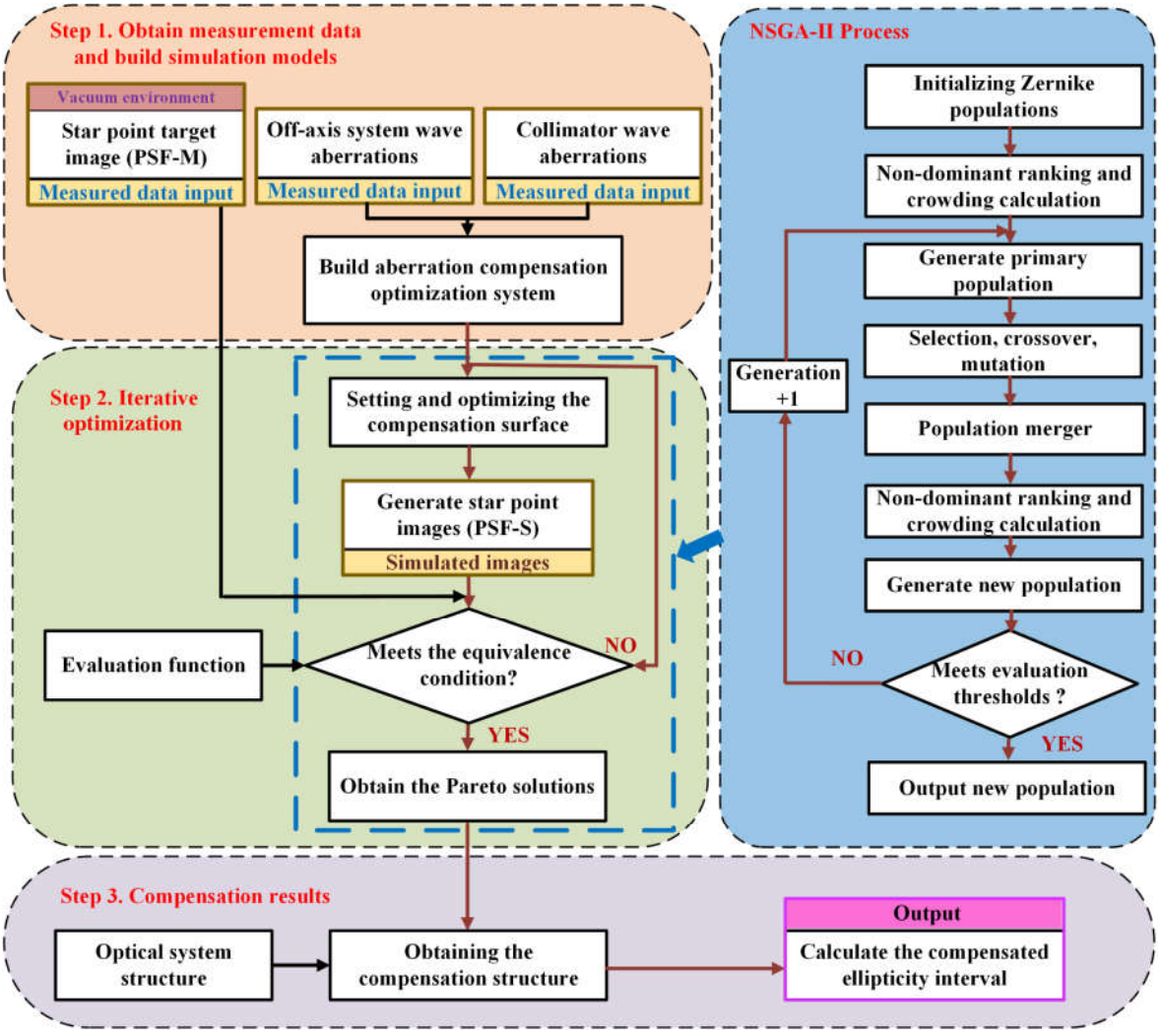


Fig. 7. Flowchart of the optimization algorithm.

combined with the contents of Section 2.E. The acquisition of optimization results can be started with the training section and NSGA-II parameter settings, respectively.

4. SIMULATION RESULTS

In this section an example is first shown to demonstrate the flow of the optimal compensation method, including optimizing the setting of the target PSF, introducing random errors, and obtaining compensation results. Then, the robustness of the algorithm is verified by setting multiple sets of cases with different error effects for the same optimization objective and different optimization objectives.

A. Simulation Settings

In order to verify the efficacy of the method in this paper, it is necessary to obtain an ideal PSF with a known PSF ellipticity. The PSF of an off-axis optical system generated by an optical design software is used as the target (PSF-M), which has a PSF ellipticity value of 0.0349, as shown in Fig. 8(a). In order to simulate the effects of wavefront measurement errors and flat mirror on PSF ellipticity, the process of setting the error is given. The wave aberrations of the collimator and the off-axis optical system are measured for the same FOV. In Fig. 2(b), the measurement errors (3σ) are counted, while the error range of Zernike coefficients (Z4–Z9) are listed in Table 1. It is added to the target as simulation error.

In order to simulate the wavefront measurement errors of the 2 m aperture optical system [mainly originating from items Z4–Z9, as shown in Fig. 2(b)] and the effects of the flat mirror, random errors are added to the target. As shown in Eq. (11), ε_i is a normal distribution with $\bar{R}_i$ as the mean and σ_R as the variance, $\varepsilon = \{\varepsilon_{Z4}, \varepsilon_{Z5}, \cdots, \varepsilon_{Z9}\}$. $\bar{R}_i$ indicates the average of Zernike coefficients for Z4–Z9 terms, respectively. The σ_R indicates 3 times the standard deviation of the Zernike coefficient, respectively. A normally distributed random number was added to each Zernike coefficient of the target PSF as the measurement error. As shown in Eq. (12), c_i indicates the Zernike coefficient

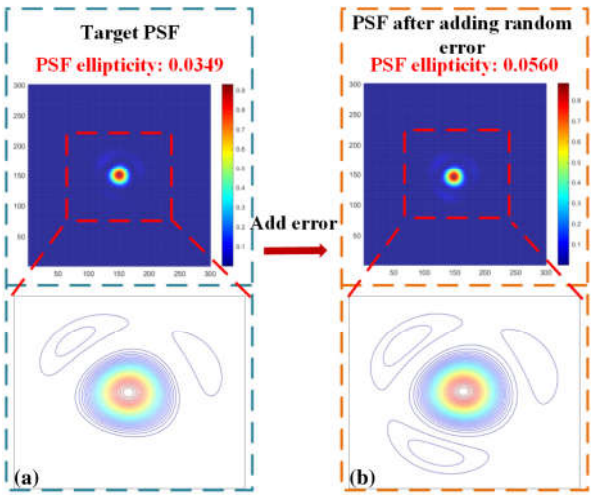


Fig. 8. Optimization objective and error setting. (a) Target PSF shape distribution. (b) Error PSF shape distribution with Zernike coefficients.

Table 1. Error Range of Zernike Coefficients (Z4–Z9) R_i

	Numerical Range (3σ)
Defocus (Z4)	–0.15–0.01
Astigmatism, Primary (axis at 0° or 90°, Z5)	–0.07–-0.03
Astigmatism, Primary (axis at ±45°, Z6)	0.03–0.06
Coma, Primary (x axis, Z7)	0.03–0.06
Coma, Primary (y axis, Z8)	0.02–0.09
Spherical Aberration, Primary (Z9)	0.03–0.05

after adding the error, and a_i indicates the Zernike coefficient of the target PSF. After adding the errors, the target PSF produces a large side flap, as in Fig. 8(b), and the ellipticity value becomes 0.0560. Next, a detailed description of the optimization process is provided in the subsequent section:

$$\varepsilon_i \sim N(\bar{R}_i, \sigma_R^2), \tag{11}$$

$$c_i = a_i + \varepsilon_i. \tag{12}$$

The initial settings of NSGA-II are as follows: the initial population number is 180, the maximum number of iterations is 120, the genetic crossover parameter is 1, and the probability of variation is 0.1. In the optimization iteration process, the PSF ellipticity, the EE80, and the PSF morphology parameter are the three mutually independent evaluation functions that have the same effect on iterations.

B. Simulation Process and Results

The processes of variation in the differences between evaluation values and target values over iterations are illustrated in Figs. 9(a)–9(c). The x axis represents the number of iterations, while the y axis represents the logarithm of the evaluated value. As the number of iterations increases, the evaluation function decreases significantly, and the iteration curve gradually converges. By assigning weight coefficients of 0.5 to w_1 and w_2, and 1 to w_3, w_4, and w_5, the convergence process of the comprehensive evaluation function illustrated in Fig. 9(d) is depicted. After 17 iterations, the evaluation value is reduced to within 0.01. When the evaluation value of the PSF morphology evaluation function is less than 10^{-3}, the subgroup is obtained, meeting the required level of computational precision. Combining Figs. 9(a)–9(d), it can be concluded that for the model described in Section 3.A, significant optimization can be achieved by the evaluation function set up in this paper.

To illustrate the process of calculating the PSF ellipticity compensation values for the off-axis optical system, the distribution of the 42nd generation subgroup is plotted, as shown in Fig. 10. Each ball represents a potential solution. The Zernike coefficients of each potential solution, which correspond to the compensation surface, are extracted from the population and substituted into the optical design software to construct the compensation model. The PSF ellipticity value e, long-axis value e_1, and EE80 are calculated for each potential solution. Four solutions from the 120 potential solutions are selected as illustrations and are represented as balls of different colors in Fig. 10. The four solutions are analyzed in the following paragraph.

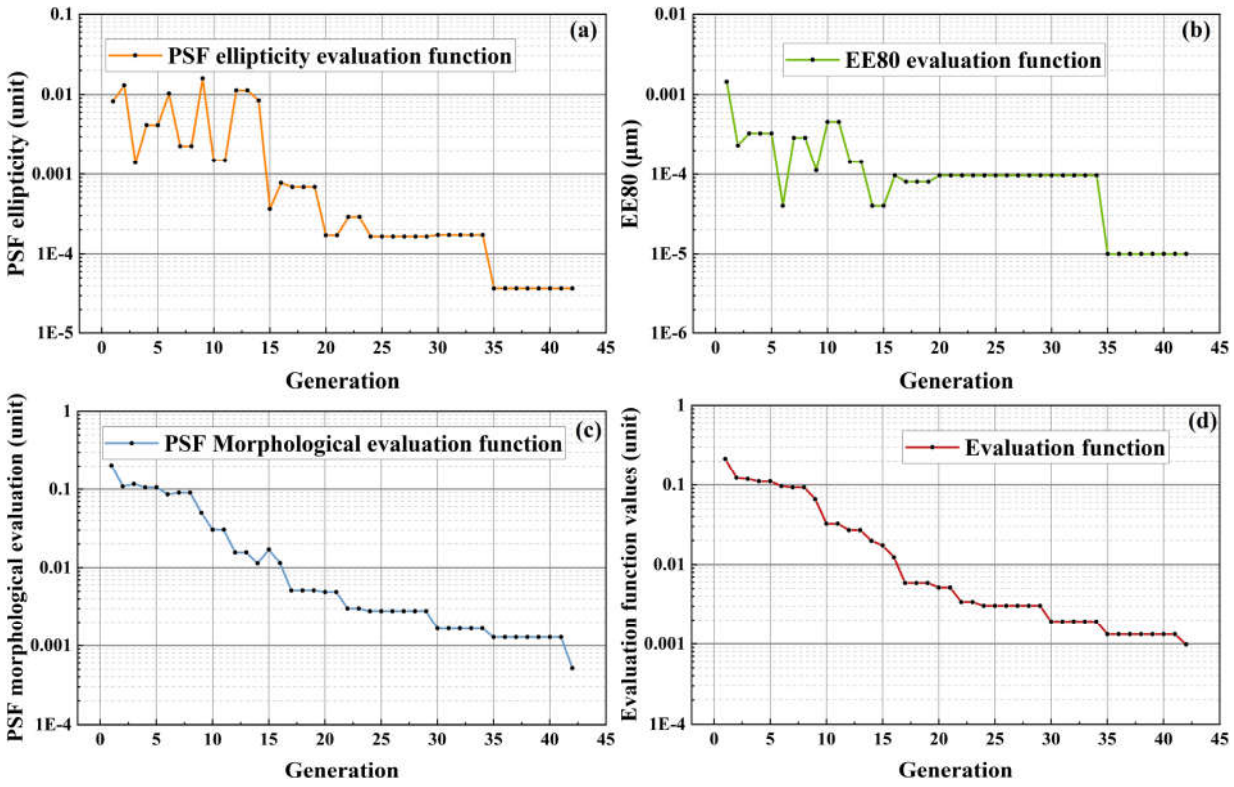


Fig. 9. Convergence curves of each evaluation function. (a) PSF ellipticity, corresponding to Eq. (8). (b) EE80 radius function, corresponding to Eq. (9). (c) PSF morphological function, corresponding to Eq. (10). (d) Integrated evaluation function.

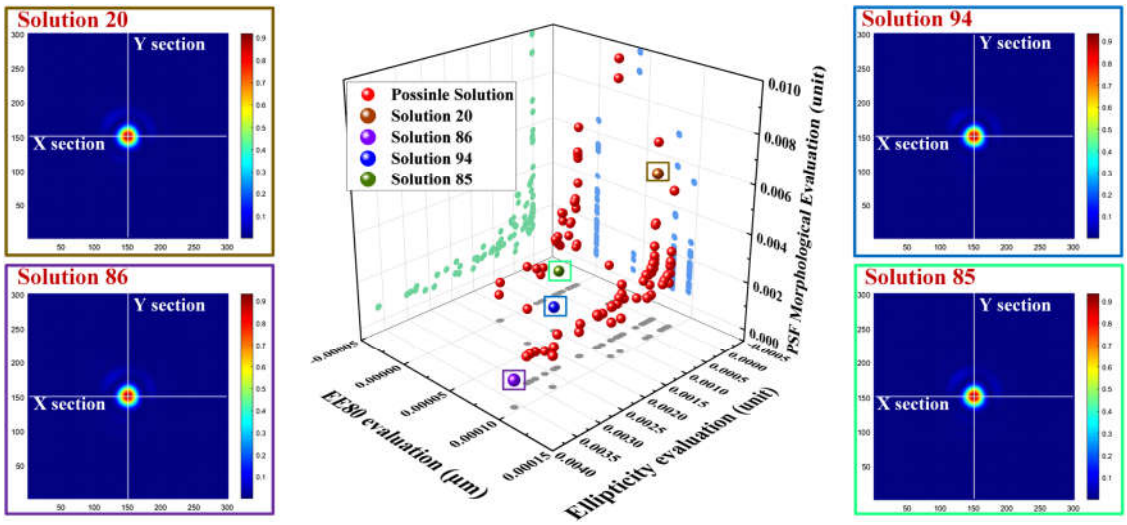


Fig. 10. Distribution of the most optimal subgroups.

The intensity curves of the X axis and Y axis of the four solutions are plotted separately, as shown in Figs. 11(a)–11(d). In order to facilitate a clearer comparison of the differences between the solutions, the parameters of the four solutions are calculated separately, as shown in Table 2. The ellipticity evaluation value of solution 20 (orange ball) is the smallest in the subgroup, and it has a better EE80. Distinctively, the ellipticity evaluation value of the solution 86 (purple ball) is the maximum value in the subgroup. The ellipticity evaluation value and EE80 of solution 94 (blue ball) are moderate. Solution 85 (green ball) has the best PSF ellipticity and EE80 deviated from the target values by 4×10^{-4} and 1×10^{-4}, which is the pareto solution among the potential solutions. To demonstrate the differences in PSF before and after optimization, solution 85 was used to compare with the target PSF and the PSF after adding the error, as shown in Fig. 12. Obviously, in terms of PSF shape and

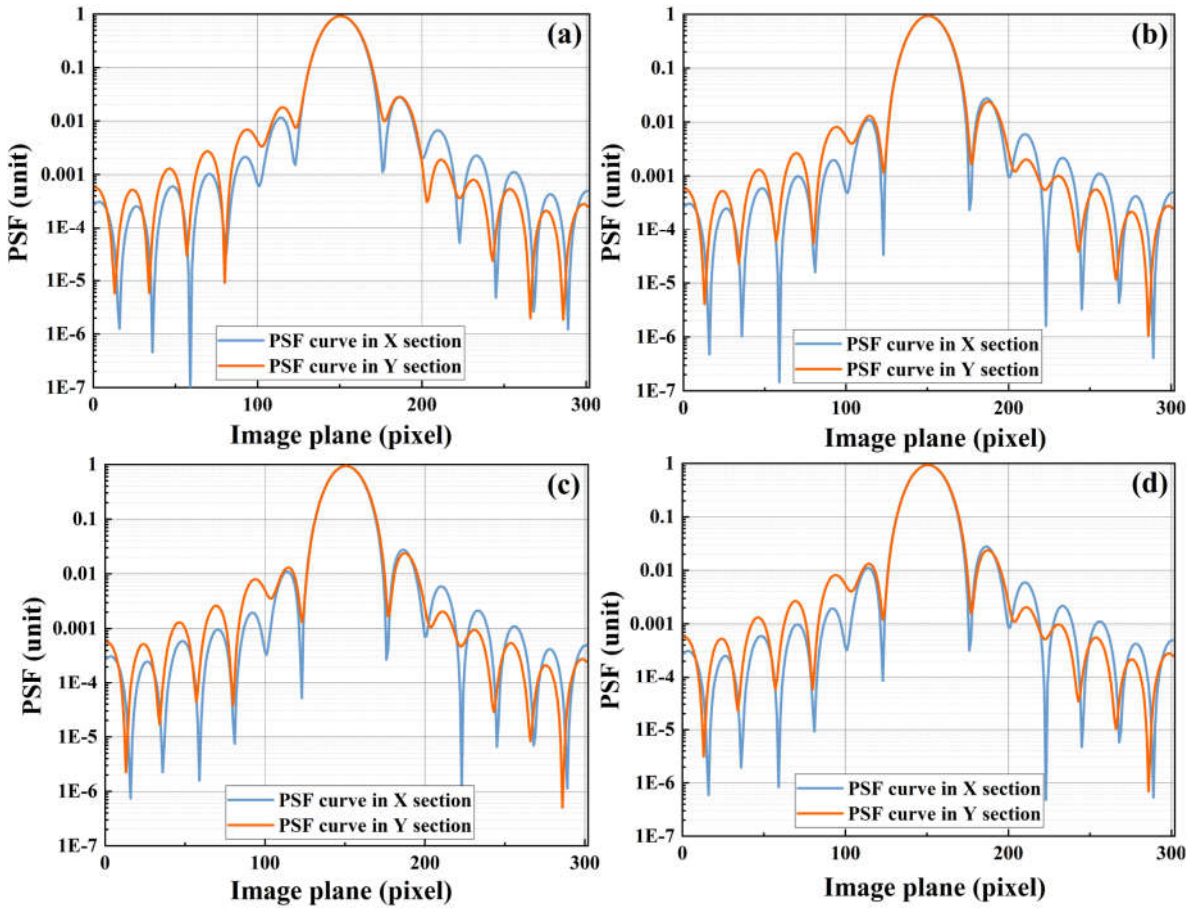


Fig. 11. Cross-sectional and vertical curves of the four solutions of PSF. (a) Solution 20. (b) Solution 86. (c) Solution 94. (d) Solution 85.

ellipticity, solution 85 is closer to the target PSF than the PSF after adding errors.

Since there is an incomplete correspondence between PSF ellipticity and PSF, the four solutions filtered by the composite evaluation function $F_c(f_e, f_r, f_p)$ can only represent special cases in the optimization subgroup. However, the purpose of this paper is to accurately measure the PSF ellipticity e of an optical system. Therefore, all potential solutions should be counted, as shown in Fig. 13. The target value, upper and lower 95% confidence interval deviations, and initial values were calculated and annotated in Fig. 13. The target value of PSF ellipticity e is 0.0349, and the compensation distributed between 0.0344 and 0.0356. The target value of long-axis e_1 is 0.0328, and the compensation distributed between 0.0321 and 0.0331. The compensation intervals include the target value, and the range of intervals is less than 0.01. Evidently, the method described in this paper effectively restores the PSF which encompasses wavefront measurement errors and the effects of flat mirror, enabling high-precision measurement of PSF ellipticity.

Table 2. Evaluation Values of the Four Solutions and the Target Solution

	PSF Ellipticity (e)	e_1	f_r (μm)
Target value	0.0349	0.0328	0.0262
Initial value	0.0560	0.0511	0.0291
Solution 20 (orange ball)	0.0251	0.0233	0.0260
Solution 86 (purple ball)	0.0393	0.0352	0.0252
Solution 94 (blue ball)	0.0340	0.0313	0.0250
Solution 85 (green ball)	0.0345	0.0330	0.0261

C. Robustness Verification

In order to validate the compensation effect of the algorithm proposed in this paper under different error conditions, more simulations are performed for various scenarios. These share consistent genetic parameter settings, and their optimization results are illustrated in Fig. 14. Figure 14(a) shows the shape distribution of the target PSF. Figures 14(b1)–14(b4) show the same target PSF under four different error conditions. Figures 14(c1)–14(c4) show the pareto solutions after the corresponding compensation recovery. Obviously, all the four different PSFs shown in Fig. 14(b) are optimized to be the same as the target PSF, which indicates that the method proposed in this paper is effective even though error distributions are different. For the PSFs with four different error conditions, the solutions screened in the optimal subgroup all well compensated. Compare the PSF ellipticity values before and after

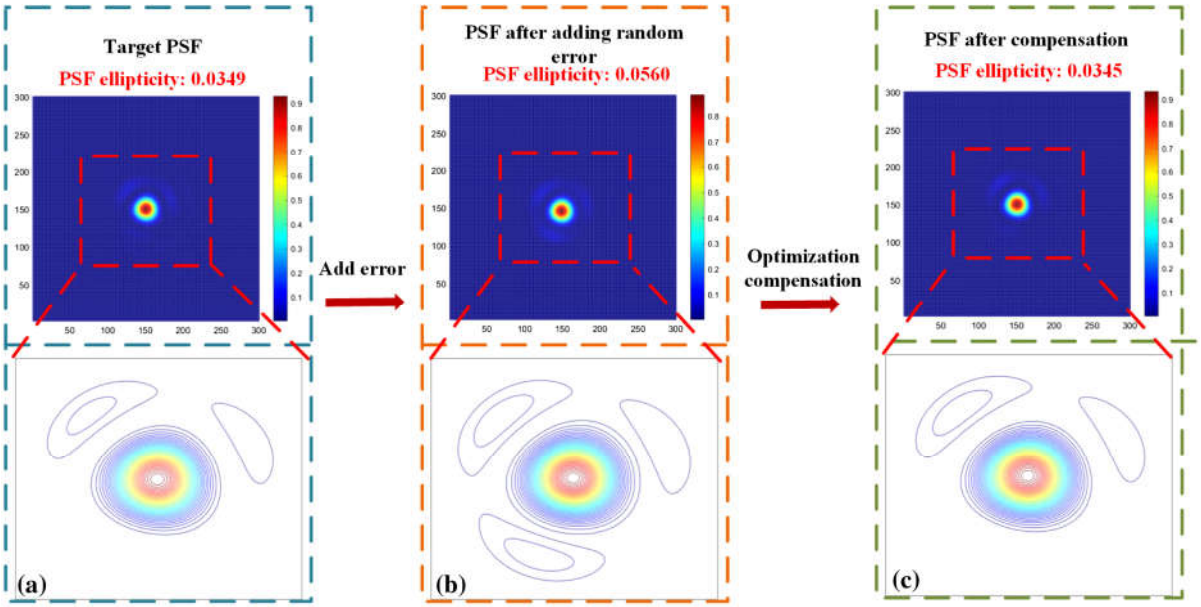


Fig. 12. Optimization process. (a) Target PSF shape distribution. (b) Error PSF shape distribution. (c) Solution 85 in the optimal subgroup.

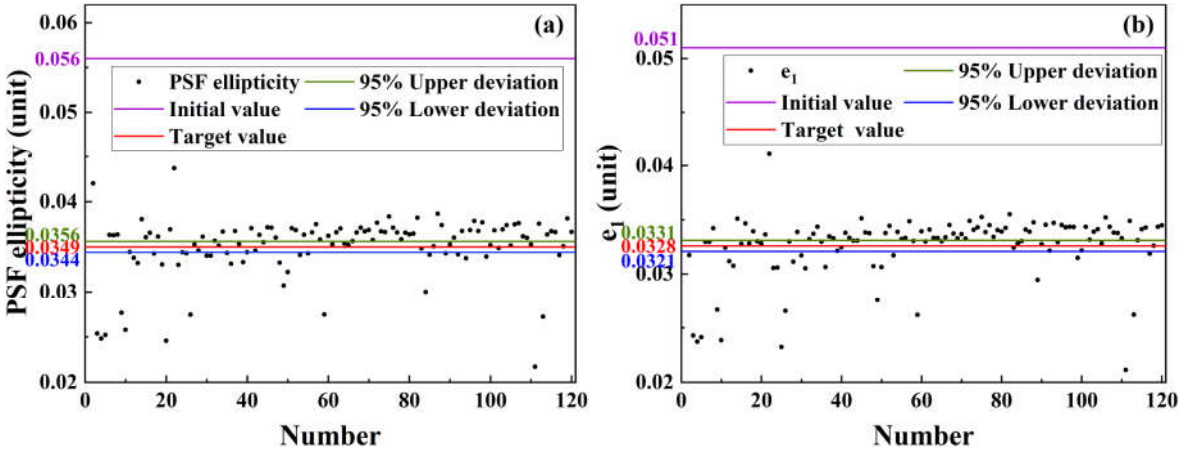


Fig. 13. Distribution range of all potential solutions. (a) PSF ellipticity values. (b) Long-axis values.

compensation, as shown in Fig. 15. The yellow and purple bars are the distribution intervals (σ) of all the compensated solutions, and the green bar is the initial value before compensation. Analyzing and comparing each set of data, the compensation intervals included the target ellipticity value and the range of intervals was less than 0.01. Combined with Fig. 14 and Fig. 15, for different optimization objectives, good compensation results are obtained through optimization iterations. It can be seen that the method in this paper has good optimization results and performance for the different target PSFs.

In the previous situations, the target PSF was the same. Here, the target PSF is altered to verify that the method proposed in this paper can be applied to different wave aberrations cases. The new target PSFs are shown in Fig. 16, which is quite different from the target PSF shown in Fig. 14(a). The PSF ellipticity value of the new target PSFs are shown with the orange bar in Fig. 16. As described in Section 4.A, errors are introduced into the target PSFs, as illustrated by the green bar. The distribution intervals (σ) of all the compensated solutions in the optimal subgroup are obtained, as illustrated by the yellow and purple bars, respectively. For the PSFs with different error conditions, the solutions screened in the optimal subgroup are all well compensated. Analyzing and comparing each set of data, the compensation intervals included the target ellipticity value and the range of intervals are less than 0.01. In summary, for different optimization objectives and different error cases, significant compensation recovery results can be obtained by the optimization algorithm and evaluation function, which demonstrates that the method proposed in this paper is correct and effective.

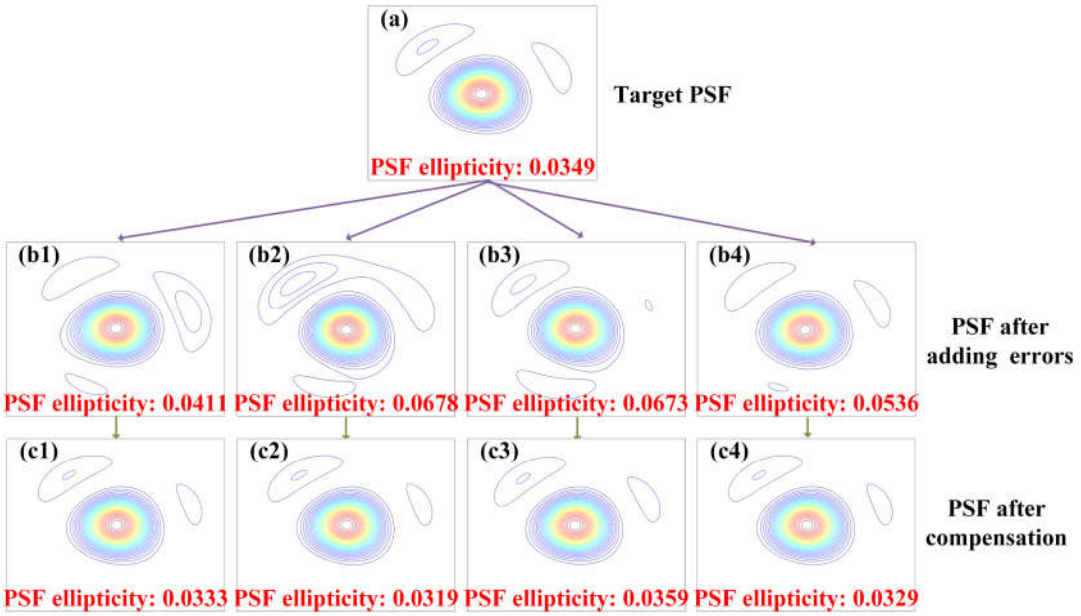


Fig. 14. PSF compensation recovery results. (a) Target PSF shape distribution. (b1) to (b4) PSF shape distribution with errors. (c1) to (c4) PSF shape distribution of the pareto solution after compensation.

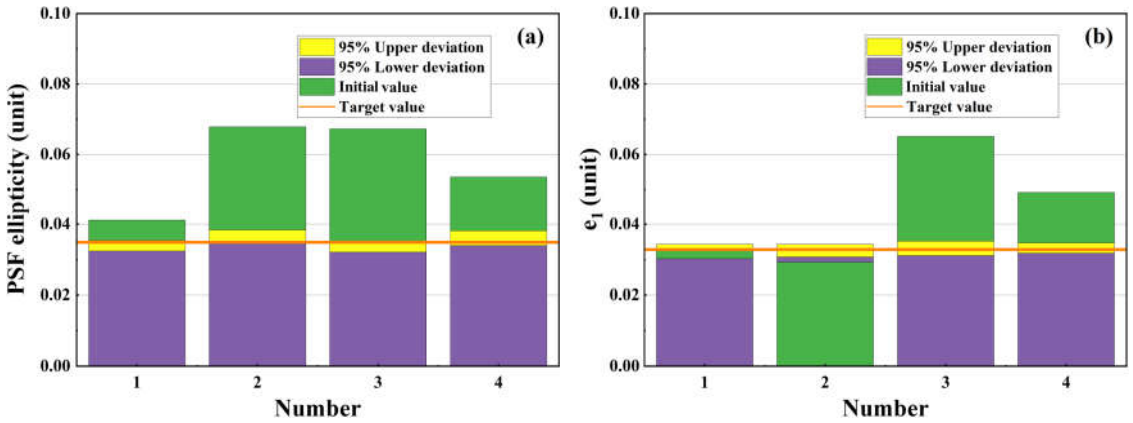


Fig. 15. Compensation calculation results for different error cases. (a) PSF ellipticity values. (b) Long-axis values.

5. DISCUSSION AND CONCLUSION

PSF ellipticity of telescopes depends on many factors, such as optical design, manufacturing errors, optical alignment, and so on. Hence, there may be large differences between the actual PSF ellipticity of telescopes and those predicted in their designs. In this paper, the research object is an off-axis telescope that would be used to detect WGL in space. According to the error distribution budget of the telescope, the PSF ellipticity measurement error should be smaller than 0.01. To ensure that the optical properties of the telescope allow accurate WGL detection, it is necessary to measure PSF ellipticity of the telescope on the ground, in which collimators are indispensable. Based on the characteristics and shortcomings of the two common methods, i.e., the wave aberration numerical calculation method and star point target imaging method, a multi-objective optimization method is proposed to reduce the effects of the main measurement errors on PSF ellipticity of the telescope. The following conclusions can be drawn:

1. Due to the large aperture and long focal length, it is difficult to precisely measure the wave aberrations of the telescope. In addition, a 2 m flat mirror, which is indispensable to measure wave aberrations of the telescope, has significant surface errors. Hence, the wave aberration numerical calculation method cannot meet the requirements of PSF ellipticity measurement of the off-axis space telescope. By analyzing the errors of the wavefront measurement experiment, it is found that typical first-order astigmatism (Z5–Z6) on PSF ellipticity affects 0.034, and the first-order coma (Z7–Z8) affects 0.012. As for the star point target imaging method, a collimator is indispensable and wave aberrations of the collimator induce PSF ellipticity variations to be 0.0296. Obviously, both methods cannot

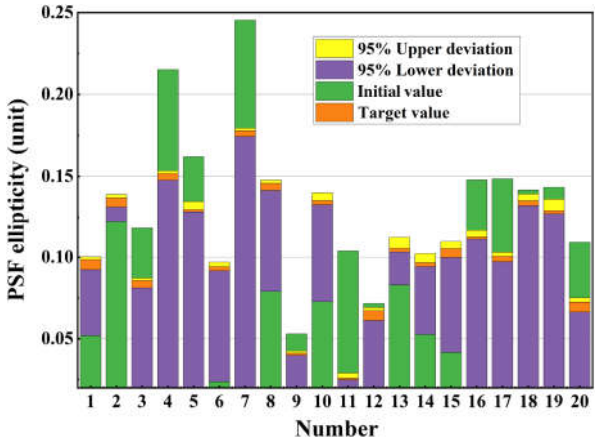


Fig. 16. PSF ellipticity compensation results for different targets.

achieve the requirements of PSF ellipticity measurement accuracy.

2. A multi-objective optimization and compensation method is proposed to obtain the PSF ellipticity of the off-axis telescope. To compensate for the effects of the wave aberrations introduced by the collimator, an equivalent model is constructed. NSGA-II and three different evaluation functions are involved to build and optimize a "virtual lens" that is used to compensate for wavefront measurement errors and the effects of flat mirrors. The three evaluation functions include PSF ellipticity evaluation function, EE80 radius function, and PSF morphological function.
3. Via a specific optimization example, it is proved that the method proposed in this paper is correct and feasible. According to the measured wave aberrations of the telescope, its PSF is calculated via the Fourier transform and treated as the target PSF whose ellipticity is 0.0349. To simulate the wave aberrations of the collimator, wavefront errors are added into the optical system via simulated soft, and a new PSF whose ellipticity is 0.056 is obtained. It can be seen that the effects of the wave aberrations of the collimator on PSF ellipticity are significant. After optimization via the method proposed in this paper, the PSF ellipticity of the optimized system becomes 0.0345, which is very close to the target PSF whose ellipticity is 0.0349. Obviously, the wave aberrations of the collimator are compensated correctly, and good performance is achieved.
4. In order to check the robustness and universality of the method proposed in this paper, different wavefront errors and different target PSFs are involved. Excellent compensation recovery results are obtained. The compensated intervals all contain the target value, and the range of the intervals is less than 0.01, which demonstrates that the method proposed in this paper is effective and robust.

The method proposed in this paper can be widely applied to PSF ellipticity measurement of optical systems and has good engineering application potential. Currently, the method has a relatively long running time (about 6 h per set) due to the high computational complexity. In the future, it is necessary to construct more appropriate optimization evaluation functions so that the iterative procedure can converge faster.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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